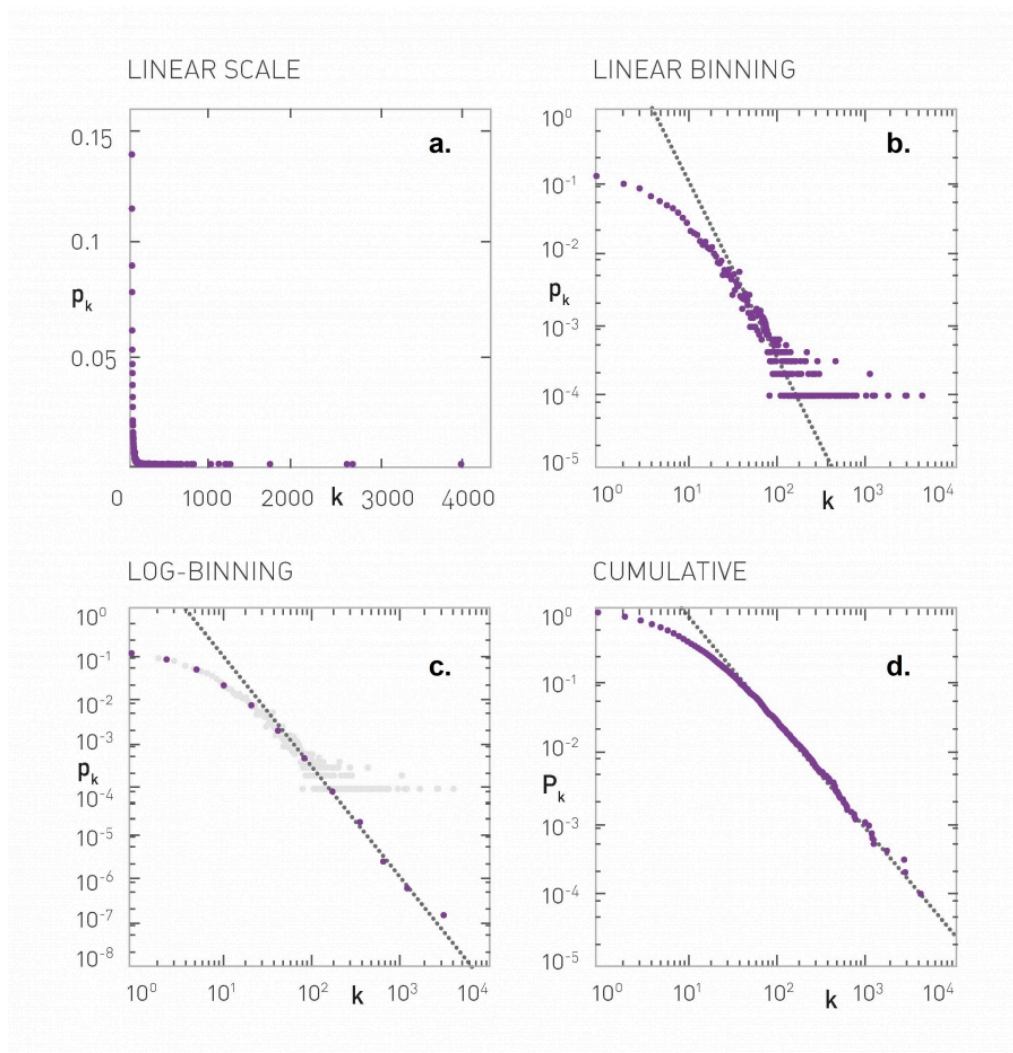


L4: How to Plot Power-law Degree Distribution



Plotting a Degree Distributions, Image 4.22 from networksciencebook.com by Albert-László Barabási

There are several ways to plot a power-law distribution – but not all of them are good and some of them can even be misleading.

This plot visualizes the same distribution, $p_k = c'(k + k_0)^{-\alpha}$, with $k_0 = 10$ and $\alpha = 2.5$, in four different ways. The term $k_0 = 10$ causes the “low-degree saturation” that you see at the left tail of the distribution, decreasing the probability of lower-degrees.

The four plots are:

- linear-linear scale. This is clearly a bad way to plot a power-law distribution.
- log-log scale but with linear binning (the bin width of the histogram increases linearly with k). This is also not an appropriate approach because the bins of the histogram for high values of k are quite narrow and they often include 0 or 1 measurements, creating the “plateau” that you see at the tail of the distribution.
- log-log scale but with logarithmic binning (the bin width of the histogram increases exponentially with k). Note how the tail of the distribution drops almost linearly with k when the degree is higher than about 50-100. A potential issue with this approach is that we still need to figure out how fast to increase the bin width with k .
- the Complementary Cumulative Distribution Function (C-CDF), which shows the probability P_k that the degree is higher or equal than k (all previous plots show the probability p_k that the degree is equal to k). This is the best approach because

we do not need to determine an appropriate sequence of bin-widths. Please note however that the slope of the C-CDF is not the same as the slope of the degree distribution. In this example, $\alpha = 2.5$ and so the exponent of the C-CDF is 1.5.